



The continuity and change in mega-urbanization in Indonesia: A survey of Jakarta–Bandung Region (JBR) development

Tommy Firman*

Research Cluster on Regional and Rural Development, School of Architecture, Planning, and Policy Development, Institute of Technology, Bandung 41032, Indonesia

A B S T R A C T

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The physical development of the Jakarta Metropolitan Area (JMA) and the Bandung Metropolitan Area (BMA) is characterized by the formation of an urban belt from Jakarta to Bandung of about 200 km, reflecting a growing mega-urban region marked by a mixture of rural and urban activities and blurred rural–urban distinction. This process is continuing at an even with faster rate than in the period from the 1980s to the mid-1990s. In fact, the JMA and the BMA are both experiencing transformation from single-core to multi-core cities. In the past three decades the development of socioeconomic activities has resulted in several environmental problems. The JBR also faces problems of uncontrolled land conversion. New town developments have reinforced spatial segregation by life style and income status while industrial estates are also growing in the region. There is a tendency of local governments in the JBR to try to exploit local resources excessively. The JBR is a functional region that should be managed as an inseparable region through an integrated spatial development plan. Institutional development is now critical to ensure the sustainability of the JBR.

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Introduction

The contemporary urbanization in Asia indicates the blurring distinction between 'rural' and 'urban' (Brennan, 1999; Hugo, 2006; Jones, 2002, 2006; McGee, 1995). Urban development in several Asian countries is characterized by physical growth which extends beyond metropolitan and city boundaries, and radiates from city centers in all directions (Hugo, 2006; Jones, Tsay, & Bajracharya, 1999; McGee & Robinson, 1995). The development is also marked by many different economic activities and land uses, including industrial estates, large-scale housing projects, and tourist resorts, interspersed with existing agricultural activities (McGee & Robinson, 1995), which is referred to as the phenomenon of mega-urbanization (Firman, 2003; Hugo, 2006; Jones, 2002, 2006; Lin, 1994; McGee, 1995). The policies of many Asian countries to control the sprawl and expansion of metropolitan areas and to slow rural to urban migration have been simply without a great success if not total failures.

A study of extended metropolitan region of Jakarta (Jabotabek) and Bandung (BMA) in the early 1990s (Dharmapatni & Firman, 1995; Firman & Dharmapatni, 1995) concluded that Jabotabek and BMA are physically being integrated, shaping a belt of an urban region from Jakarta to Bandung (Fig. 1), reflecting the formation of

mega-urban. This urban belt is characterized by a mixture of socioeconomic activities, including agriculture, industries, trade and residential, which in turn has created very intense rural–urban linkages, blurred the rural–urban distinction and made very distinct settlement patterns (Firman & Dharmapatni, 1995: 181–3).

There are several factors contributing to the process of mega-urbanization in Jakarta–Bandung Region (JBR), most notably large-scale housing and new town, infrastructure and industrial estate development. Development of this region has greatly enhanced agglomeration economies. As in many Asian countries, recent urbanization and urban development in Indonesia is greatly triggered by domestic and direct foreign investment, notably in the sector of industry and services (see Firman, 2002), and more recently by infrastructure development. Those economic activities tend to locate in large cities, notably Jakarta, Surabaya, Bandung and Medan, due to 'urbanization economies', including the availability of infrastructure and utilities, such as water supply, electricity; seaports and airports; and concentration of skilled labor and markets.

Against the above background, the present study will examine the continuity and change in the process of mega-urban formation in JBR, which is the largest concentration of urban population and economic activities in Indonesia and one of the largest in Southeast Asia. There are several factors that might have affected the region (JBR) during the last decade, notably the economic crisis at the end of 1990s followed by political reformation and the new policy of regional autonomy and fiscal decentralization in Indonesia which

* Tel.: +62 22 2504735; fax: +62 22 2501263.
E-mail address: tfirman@melsa.net.id

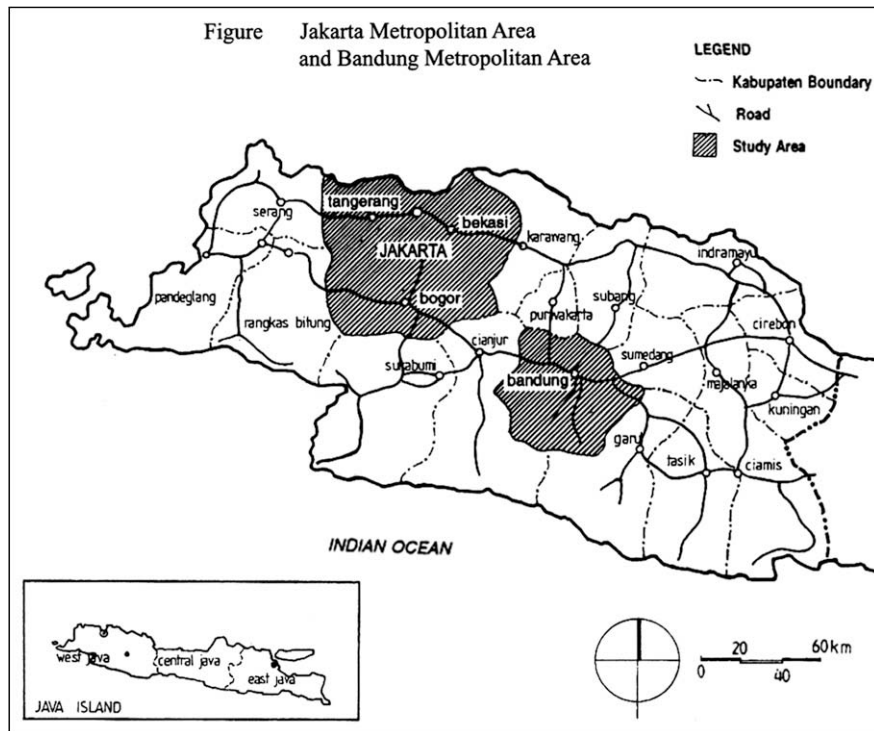


Fig. 1. Jakarta Metropolitan Area and Bandung Metropolitan Area (source: Dharmapatni & Firman, 1995, p. 297).

has been started since 2001. This study is not particularly intended to enter the debate over whether the mega-urbanization in Southeast Asia, such as Indonesia, resembles that in western countries (see Dick & Rimmer, 1998; McGee, 2005), but rather aimed to examine the process of mega-urbanization within the context of Indonesia's recent socioeconomic development and political conditions. The study argues that first of all mega-urbanization in Southeast Asia needs to be understood in the contexts of the broader socioeconomic, political situation, as well as the physical condition of the places and societies where the region exists, not on whether the urbanization in developing countries resemble or differ from western countries as such. As Shatkin (2007: 2) correctly maintains that there is a need to shift the focus in study of urbanization at global level from 'world city model' towards a more grounded examination of the interaction between global and local actors and institutions in a particular setting. The present study intends to make contribution in that perspective.

There are two occurrences which might affect the recent urbanization pattern in Indonesia. The first is the economic crisis at the end of 1990s and early 2000s, whereas the second is the new policy on regional autonomy and fiscal decentralization. The recent economic crises have had a big impact on urbanization and urban development in Southeast Asia (see Chatterjee, 1998), most notably Indonesia. Economic growth had seemed to significantly contribute to urban development in Indonesia from the late 1970s until the mid-1990s, but the economic crises made it uncertain. As the economic activities in JBR slowed down, the number urban unemployed increased rapidly during the crises, while a substantial number of workers in Jakarta Metropolitan Area (JMA) and Bandung Metropolitan Area (BMA) have involved in many marginal jobs, such as itinerant food traders, just for survival reasons.

Indonesia's new regional autonomy and fiscal decentralization reform could potentially change pattern of urban and regional development (see Firman, 2003). The question to address in this study is to what extent have the regional autonomy and fiscal decentralization reforms affected the development of Jakarta–

Bandung Region (JBR)? Analyzing impacts of the decentralization reform on the development of the region in Indonesia could be a premature task, as the implementation of this reform since 2001 is still underway, while 7 years (2001–2008) is too short a time to examine the impacts. Moreover, identifying the impacts of this reform will also be a formidable task, as the reform has broad socioeconomic, culture and political dimensions. However, given the process of such reform as described above, it is possible to analyze the potential impacts of the reform on urban and regional development, including the Jakarta–Bandung Region (JBR).

The new regional autonomy and fiscal decentralization policy implies that urban and regional development in Indonesia, including in the Jakarta–Bandung Region (JBR), have become local authority affairs, meaning that it has to be planned and implemented according to local needs by local government and communities. In the past urban and regional development in Indonesia had been strictly under the control and direction of central government through presidential grant allocation (Inpres) and ministerial programs.

The Jakarta–Bandung Region (JBR) basically comprises the two largest concentration of urban population and urban economic activities in Indonesia, namely Jakarta Metropolitan Area (JMA) and Bandung Metropolitan Area (BMA). JMA, is also now called "Jabodetabekjur", an acronym which stands for Jakarta–Bogor–Depok–Tangerang–Bekasi–Cianjur. This region which is located in the northern area of West Java (Fig. 1), comprises of 12 administrative units at different levels: First, the Jakarta Special Region (DKI Jakarta) having provincial government status; second, eight municipalities (Kota) and districts (Kabupaten), namely the Municipalities of Bogor, Depok, Tangerang and Bekasi, and the Districts of Bogor, Tangerang, Bekasi, and Cianjur. Although the JMA comprises of only 0.33% of the national land area, it accommodates as much as 12% of Indonesia's total population, and produces nearly one-fourth of Indonesia's Gross Domestic Product (GDP) at present (Rustiadi, 2007). Meanwhile, the Bandung Metropolitan Area (BMA) comprises one municipality and district, namely the

Municipality of Bandung and the District of Bandung. In total, the Jakarta–Bandung Region (JBR) covers an area of more than 12 275 km², whereas Jakarta City and Bandung City occupy a relatively small area, that is, only 164 and 81 km², respectively.

The total population of the JBR was about 32 million in 2004, while the population of Jakarta City as the core of the region was about 8.5 million, which was 26.5% of the total population, whereas Bandung City had a population of about 2 million. The population density in the region is highly uneven, in which the municipalities (Kota) have much higher population density than the districts (Kabupaten). Jakarta City is the densest with 45.6 persons per km², whereas District (Kabupaten) Cianjur is the lowest with only 0.59 persons per km².

The primacy index, i.e. the proportion of urban population in Jakarta Metropolitan Area (JMA) relative to the four largest metropolitan regions in Indonesia, including Jakarta, Surabaya, Bandung and Medan was 0.56 in 1980, 0.58 in 1990, and 0.57 in 2000 (see [Firman, 1997](#)). This clearly reflects the Indonesia's substantial urban population concentration in the JMA. In fact, the JMA's share of the national urban population reached 22.5 in 1980, 23.6% in 1990, and 21.2% in 2000.

While population plays a very important role in development of Jakarta–Bandung Region (JBR), commuting is also evident. Millions of people commute daily between Jakarta City and the adjacent areas, including the Kabupatens (District) and Kotas (Municipality) of Bogor, Tangerang, Bekasi, by bus, private cars and trains. According to Japan International Cooperation Agency (JICA) (see [Hidayat, 2007](#)) by 2002 alone there were more than 3 million commuters between Jakarta and its surrounding areas, including 1.10 million between Tangerang and Jakarta; 1.14 million between Bekasi and Jakarta; and nearly 1.00 million between Bogor and Jakarta. This figure does not include the commuters for farther areas, such as Cianjur, Serang, Rangkasbitung, and Sukabumi. Likewise a substantial number of people commute between Bandung City and its surrounding areas.

The remainder of this article will be divided into five parts. The next part will discuss the mega-urbanization process in Asia to provide a theoretical context and argument for this study; The following part will analyze impacts of the economic crisis on the JBR development; There follows an examination of the current socioeconomic and physical development in the region, including urban population and employment, new town and industrial estate development, and land conversion. The next part will discuss the impacts of new Indonesia's decentralization policy on the development of the region. A final section will conclude the discussion.

Mega-urbanization in Asia

Globalization of the economy, that is, the process of deepening and expansion of the global market for commodities, finance and services, greatly supported by the fast development of transportation and communication technology and latter triggered by the trade liberalization ([Cho, 1997, 1999](#)), has greatly affected the pattern of urban development in the world. Moreover, the establishment of the WTO (World Trade Organization) in 1995, which essentially removed trade barriers, has been an impetus for export and import expansion among countries, and therefore among the large port cities in the world (see [Mera, 2002](#)). As a result, large cities in the world, including those in the developing countries, have been integrated into the global economy (see [Lo & Marcotullio, 2000](#); [Lo & Yeung, 1996, 1998](#); [Mahbubani, 2007](#); [Olds, 1995, 2002](#); [Taylor, 2004](#)). As [Ohmae \(1996\)](#) argues, globalization has greatly reduced the importance of national territorial boundary.

Many big cities serve as a production site for finance and the other leading industries, and therefore play the role as center for

control of global capital ([Sassen, 1997: 1–2](#); [Sassen, 2001](#); see also [Beaverstock & Boardwell, 2000](#)). Many scholars argue that the transnational capital has largely affected the spatial patterns of urban development in Asia ([Cho, 1997](#); [Lin, 1994](#); [Webster, 2001](#); [Zhu, 2000](#)).

As [Douglass \(2000: 237\)](#) argues that the globalization of finance, trade, and production has spurred the development of mega-urban regions in Pacific Asia and has resulted in urban spatial restructuring, which involves the polarization of a few large urban centers; formation of large mega-urban around the centers; and slowed down the urbanization rate in inland regions, especially the densely rural agricultural regions away from the city centers. More specifically, the urban restructuring in Asian Mega-Urban Region during 1990–2000 was characterized by at least seven features ([Firman, 1998: 233](#)). First, development of economic activities at a global scale; second, division of function between the core and the outskirts of the city; third, changing from a single-core to multi-core urban areas; fourth, land use change in the city center and farmland conversion in the outskirts; fifth, large-scale urban infrastructure development; great increase in space production; and seven, considerable growth of commuters and increase in commuting time.

The mega-urbanization process in Asia is reflected by increasing flows of direct foreign investment, development of transportation and communication and the growing commercialization and diversification of agricultural activities ([Lin, 1994](#); [Firman, 2003](#)). According to [McGee \(2005\)](#) the process of mega-urbanization in Asia should be seen in the context of a nexus of local–global and rural–urban relationships for two fundamental reason: First, the urban physical expansion generally takes place into areas of intense rural activities, which is also called the peri-urban interface (see also [Webster, 2001](#)). Second, the globalization of economy has spurred the flows of capital, commodities, people and information which result in both the detachment of city core, in which it is sourcing the resources from a wider global market, and integration with its adjacent extended metropolitan region in terms of using needed resources, such as water and foods ([2001: 42–3](#)).

The phenomenon of mega-urbanization has taken place in many Asian countries, including Indonesia ([Firman & Dharmapatri, 1995](#); [Firman, Kombaitan, & Pradono, 2007](#); [Hugo, 2006](#); [Pradono, 1997](#)), China ([Leaf, 2002](#); [Lin, 2002](#); [Sit, 2005](#); [Xie, Batty, & Zhao, 2007](#); [Yang, 2005](#); [Zhu, 2000](#)), Vietnam ([Leaf, 2002](#)), Malaysia ([Wong, 2006](#)), Thailand, Taiwan and the Philippines ([Jones et al., 1999](#)), India ([Dupont, 2006](#)) and also in Latin America ([Aguilar & Ward, 2003](#)) and Africa ([Briggs & Mwamfupe, 2000](#); [Simon, McGregor, & Nsiah-Gyabaah, 2004](#)), and even in Australia ([Baum, Haynes, Gellecium, & Han, 2006](#)). Mega-urban regions in Asia, however, in many ways seem to be largely disconnected from local economic activities and produce a little in terms of impacts on development for local population, which in turn has resulted in regional disparities. The mega-urbanization process has taken place not only in primate cities and the fringe areas in Southeast Asian countries, such as Jakarta, Bangkok and Manila, but also in their intermediate cities including Chiang Mai in Thailand, Bandung in Indonesia and Cebu City in Manila ([Nas & Houweling, 2000](#)).

Similar phenomena also take place in some European countries, which is referred to as 'Post Suburbia', characterized by chaotic polycentric structures, urban decline in former city districts, post-suburban developments leading to edge city units, an accelerated social technology and high technology corridors ([Borsdorf, 2004: 13](#); see also [Phelps, Parsons, Ballas, & Dowling, 2006](#); [Soja, 2000](#)). As [Borsdorf \(2004: 13\)](#) further argues:

This ['Post Suburbia'] trend is changing the structures of the outskirts profoundly. Some new areas are much more independent than the former suburbs, but they are not as

multifunctional as the traditional city centres. So a fragmented structure of specialized outskirts is emerging. They are neither urban in terms of density, multi-functionality, urban life, nor rural in terms of agriculture, seclusion, and tradition. In many cultural landscapes the traditional questions of urban geographers “Where is the centre, where the periphery, where is the urban zone, where the rural?” has lost its meaning.

For Soja (2000: 152), urban development in the US and European countries is characterized by ‘what was once central is becoming peripheral and what was periphery is becoming increasingly central’ (see also Phelps et al., 2006: 10). However, as Phelps et al. argue, despite grown up body of literature on ‘Post Suburbia’, these have rarely been incorporated into broader analyses of urbanization (Phelps et al., 2006: 11).

Nevertheless, the mega-urbanization process in Asia is largely different from suburbanization in western countries, as the spread of the cities in Asian countries has occurred at an unprecedented level (McGee, 1991, 1995); however, as Dick and Rimmer (1998) maintain that the phenomenon of mega-urbanization is not particular to Asia, as many large cities in both the developing and developed world have undergone a similar process. Likewise, Webster (1995: 32) also argues that Southeast Asia Mega-Urban regions are likely to resemble Los Angeles in function, form and socioeconomic characteristics. Moreover, according to Soja (2000: xvii) development of Los Angeles basically indicates the generality of urbanization process in the world over the last three decades, with varying intensities and different ways. Perhaps, Dick and Rimmer’s, Webster’s and Soja’s arguments are true, but mega-urbanization should be primarily placed in the context of societies and places in which the metropolitan region exist (see also Forbes, 2004; Ginsburg, 1991), not whether the process of mega-urbanization in Asian countries is similar or different from western countries. There is a need to make comparative urban development at different settings.

Economic crises

Southeast Asian countries hit by serious economic crises, which began in Thailand in July 1997, and quickly spread to Asian countries, including Korea, Malaysia, the Philippines, and, most notably Indonesia which experienced the worst impacts, including the flow away of foreign capital; loss of investor confidence; and a serious drop its currency’s value. The economic crises in Indonesia also involved the monetary mismanagement and the offshore bad debts of a number of Indonesia’s private companies, which were not able themselves commit to the schedule of repayment (see McLeod, 1997).

The World Bank (1998: 1) points out, Indonesia is a country that achieved decades of high economic growth, poverty reduction and stability and no country with the size of Indonesia in recent history has ever suffered such a dramatic economic crises. The Indonesia’s economic crisis was basically rooted in the over-borrowing and overinvestment by corporate, banking and private sectors, which in less than 10 years had amounted to almost US \$180 billion, mostly unhedged (1998: 24). However, the crises has involved not only out of economic aspects but was also connected to bad governance, most notably nepotism, corruption and collusion, in the government under President Suharto who controlled the country for more than 30 years, that is, from mid-1960s to the end of the 1990s.

The Jakarta Metropolitan Area (JMA) was the hit hardest and experienced a big contraction of its economic growth rate, from 6.0% to 8.3% per year over the period 1987–1997 to minus 2.74% in 1998–1999 (see Firman, 1999). Moreover, the Gross Domestic Product (GRDP) of Jakarta City declined from about Rp. 69 500 thousand million in 1997 to about Rp. 52 283 1000 million in 1998,

based on the constant price of 1993 (Central Board of Statistics, 1999), which made the per capita GRDP of the city dropped from about Rp. 7.5 million to Rp. 6.1 million in 1998. However, per capita GRDP Jakarta City is reported to have reached Rp. 7.1 million in 2000, about 3.7 times the national average of Rp. 1.93 million (Hata, 2003).

The total approved domestic investment in Jakarta City declined from nearly Rp. 63 700 billion in 1997 to only about Rp. 18 900 billion in 1998. During the crises, the banking sector in Jakarta City and other big cities in Indonesia experienced difficulties in their liquidity, as many of the banks had provided excessive loans to property companies, most often of their own business groups, resulted in non-performing loan (NPL) because the crises hit the property sector badly. At the time, many developers and property firms in Indonesia, most notably in Jakarta, had over-invested using unhedged and prevailing market interest short-term loans, including offshores ones for both long-term projects of building construction and land acquisition (see Firman, 2000). In fact, the foreign loans for property project had amounted US \$3.4 billion by 1998 (Properti Indonesia, 1999), so that when the Indonesian currency (Rupiah) exchange rate to US Dollar dropped significantly, a number of developers faced a big problem, since they were not able to repay the debts. Because of this problem, the central government closed many major banks in Jakarta and other big cities in Indonesia and put them under surveillance during the early 2000s.

Since the economic activities in Jakarta and Bandung City declined during the economic crises, the revenue of the city government, which derived significantly from collections and taxes from business transactions, also declined substantially. This resulted in the undermining of urban infrastructure and public service provision by the city government as well as the private sector, most notably in public transportation which mostly used import components and spare-parts.

The economic crises had significant impacts on the Jakarta-Bandung Region (JBR). As many major developers simply could not pay back loans, and some even found themselves defaulting on their loans to foreign financial institutions and banks, a number of new towns and large-scale housing development in the fringes of Jakarta and Bandung City were slowed down, which resulted in significant abandoned land added to unutilized land which had been under acquisition of developers (see Dijkgraaf, 2000). It was reported that only 50 out of 500 new towns and large-scale housing projects in Jakarta Metropolitan Area (JMA) were in proper operation in 1999 (Properti Indonesia, 1999: 13).

Similarly, many condominium and office construction projects had been delayed and stopped, simply because of the unreasonable and sky-rocketing costs of construction. The situation then resulted in rapid unemployment increase in property and related sectors. An observer (Simanungkalit, 1999, cf. Tempo, 1999) indicates that the transactions in the property sector of Jakarta City dropped significantly from Rp. 2.6 trillion in 1996 to only Rp. 0.8 trillion in 1998 (see Firman, 2002). Likewise, industrial estates in the fringes of Jakarta Metropolitan Area dropped by 30% over the period from 1997 to 1998, since the demand for industrial land went down significantly, which resulted in about 47 000 ha of idle industrial estate land at the beginning of 1998. According to Heikila (1998: 5) the picture of the housing market situation in the JMA at the late 1990s essentially reflected a grand mismatch between demand and supply for housing, as many developers were only interested to build high-rise condominium at the city center and luxury houses in the outskirts, not the that was types of housing demanded by the market.

In short, the economic crises from 1998 to early 2000s saw a decline in physical and economic development in JBR, which was clearly reflected in slowing down in new towns and large-scale

residential areas and industrial estate development due to significant decline in demand for luxury housing and industrial land. This resulted in idle land in the fringe areas of Jakarta and Bandung City. The fact is that the economic crisis has significantly reduced the demand for land for new town and industrial estate development in the Jakarta–Bandung Region.

Recent socioeconomic and physical development

Urban population

The urban population of Jakarta–Bandung Region (JBR) reached more than 24 million in 2000, which is about 28% of the total urban population in Indonesia, at rate of growth about 3.5% over the period 1990–2000 (Table 1). The proportion of urban population in Jakarta over the total urban population in the region significantly decreased from 48% in 1990 to 34.7% in 2000.

It can also be noted that rate of urban population growth in the urban centers of the region, notably in Jakarta and Bandung City also substantially declined. The rate of urban population growth of Jakarta decreased from 3.1% over the period of 1980–1990 to only 0.16% over the period of 1990–2000, whereas the urban population growth of Bandung City declined from 3.3% over the period of 1980–1990 to only 0.40% during the period of 1990–2000. This might reflect the rapid spillover of Jakarta and Bandung to the adjacent areas.

The rate of urban population growth in Bogor City increased significantly from only 1.0% during 1980–1990 to almost 11% during 1990–2000, due to the expansion of the municipal administrative areas in the city. This is also reflected in the substantial decline of urban population growth of the District of Bogor, the adjacent areas of Bogor City, from which some the subdistricts were annexed to the city, which fell from 11.7% during 1980–1990 to only 0.46% in 1990–2000. Moreover, the separation of Depok City from the District of Bogor in the mid-1990s made the urban population growth of the District of Bogor decline even faster. Likewise, the urban population growth of the District of Tangerang and Bekasi significantly declined from 20.9% to 2.5%, and from about 20% to –1.7%, respectively, over the period, because the urban centers of these two districts were detached to become a separate municipality (Kota), namely Kota Tangerang and Bekasi in the late 1990s.

A study shows that many former residents in neighborhood within Jakarta city center moved to fringe areas of Jakarta Metropolitan Area (JMA), which reflects a functional and spatial integration of the areas into the metropolitan economy (Browder, Bohlan, & Scarpaci, 1995; see also Somantri, 2007). The Population

Census 2000 (Central Board of Statistics, 2001) notes that during the years from 1995 to 2000 about 190 000 and 192 000 Jakarta City inhabitants had moved to District and City of Bekasi and District and City of Tangerang in the fringes of the JMA respectively (see also Julianery, Widyastuti, & Santoso, 2006). Meanwhile another 160 000 of Jakarta's inhabitants moved to the District and City of Bogor and the City of Depok. This situation resulted in the decline of Jakarta City's share of the total JMA population from 54.9% in 1971 to 39.9% in 2000, whereas the share of Districts and Cities of Bekasi and Tangerang increased from 10.0% to 15.3%, and from 12.8% to 19.6%, respectively (Hata, 2003: 36).

The urban transformation in the Jakarta–Bandung Region can also be identified in the changes in the number and percentage of urban localities in the region (Table 2; see also Gardiner & Gardiner, 2006). In the Jakarta Metropolitan Area (JMA) which is also called Jabodetabek, the number of urban localities increased by more than 28%, that is from 730 to 1035, over the period from 1999–2005, whereas the proportion of urban localities over the total localities increased from about two-fifths to almost three-fifths over the same period. Similarly, the number of urban localities in Bandung Metropolitan Area (BMA) increased by about 50%, that is from 266 to 412 over the same period, resulting in the increase of percentage of urban localities from about one-third to nearly one-half.

The Population Census of 2000 indicates that the Jakarta Metropolitan Area (JMA) has been a destination for recent migrants, defined as those who had moved to any Districts and Cities in JMA during the last 5 years, that is, within 1995–2000, from many regions in Indonesia. There were, in total more than 1.35 million recent migrants in JMA as recorded by the Census, about one-third and three-tenths of them originated from Central Java and West Java, respectively. Another 15% and 9% came from Sumatra and East Java (Central Board of Statistics, 2001 in Julianery, Widyastuti, & Santoso, 2006). Meanwhile, the total number of recent in-migrants to Bandung City and Bandung District reached 315.8 thousands and 200 thousand, respectively, making the total recent migrants to BMA 515.8 thousand over that period, two-third of them were from Java (West Java Office of Central Board of Statistics, 2001 in Kompas, 2006). These indicate the attractiveness of both JMA and BMA to migrants who search for jobs in the Jakarta–Bandung Region.

Land conversion

Much has been written regarding urban development in developing world, but little is known how it has affected land

Table 1
Urban Population, Jakarta–Bandung Region (JBR), 1990–2000 (in thousands).

Administrative unit	1990	As % of total population	2000	As % of total population	Rate of growth (%)	
					1980–1990	1990–2000
Municipalities						
1. DKI Jakarta	8259.3	100.0	8347.1	100.0	3.1	0.16
2. Bogor	271.7	100.0	746.9	99.5	1.0	10.98
3. Bekasi	n.a	n.a	1622.2	97.5	n.a	n.a
4. Depok	n.a	n.a	1143.4	98.0	n.a	n.a
5. Tangerang	n.a	n.a	1325.8	100.0	n.a	n.a
6. Bandung	2026.8	100.0	2136.2	100.0	3.3	0.40
Districts						
7. Tangerang	1521.0		1940.7	69.8	20.9	2.5
8. Bogor	1923.8	51.5	2014.0	57.4	11.7	0.46
9. Karawang	339.2	22.7	660.8	37.0	7.1	6.9
10. Bekasi	1153.0	54.8	967.7	58.0	19.9	–1.7
11. Cianjur	241.8		482.6	24.8	3.0	7.1
12. Bandung	1291.9		2656.5	63.9	7.3	8.1
Jakarta–Bandung Region	17 028.5		24 043.9			3.5

Source: Central Board of Statistics, 1991; Central Board of Statistics, 2001.

Table 2

Number of urban and rural localities in Jakarta and Bandung Metropolitan Areas, 1999–2005.

Cities and districts	1999			2005		
	Urban	Rural	Total	Urban	Rural	Total
Jabodetabek (JMA)	730	1099	1829	1035	812	1847
Jakarta city	265	–	265	267	267	267
Bodetabek	465	1099	1564	768	–	1580
Bandung Raya (BMA)	266	593	859	412	451	863
Bandung city	133	6	139	154	–	154
Bandung and Sumedang	133	587	720	258	451	709

Source: Calculated from Gardiner and Gardiner (2006).

conversion, that is the process by which land is converted from agricultural to urban uses and from one urban use to another (see Firman, 2000). In many cases it may involve both the outward expansion of the physical structure and internal reorganization of urban areas, but obviously the primary reason for the conversion of agricultural land is that under urban uses a much higher rent is recovered (Grigg, 1995).

Land conversion is essentially a normal outcome of the urban development process, but in Indonesia it greatly reflects the operation of large private developers attempting to extract the highest possible rent. Moreover in large cities such as the Jakarta Metropolitan Area (JMA) and Bandung Metropolitan Area (BMA) land conversion is greatly uncontrolled due to the weaknesses of the land permit system, especially in their enforcement. Land conversion could in turn bring both direct and indirect impacts, which includes loss of agricultural land, the loss of investment in irrigation infrastructure and the influx of population into fringe areas from urban ones.

During the past three decades, the development of economic activities in the Jakarta–Bandung Region (JBR) has resulted in the extensive land conversion of prime agricultural land into non-agricultural land (see Firman, 2000), especially by new towns, subdivisions and industrial estates in the fringe areas. This could significantly affect the foodstuff production and waste of investment in irrigation of the paddy fields. Meanwhile, in urban centers many former residential areas have been converted into offices, condominiums, entertainment and business spaces. In the Jakarta CBD, a construction of Superblock Sudirman has completed recently, consisting about 2 million sq m of office, hotel, and residential spaces. Meanwhile, at present there are 16 others super-blocks in Jakarta's CBD, including Grand Kuningan, Taman Rasuna, Senayan Square and BNI City.

In Jakarta City as many as 30 big shopping malls have been built by 2006, including Ratu Plaza Mall, the first shopping mall in the city built in 1986 and the largest one Cilandak Town Square (Citos) in South Jakarta. Meanwhile, some more large shopping malls are to be built by 2008, including Gajah Mada Square, Pacific Place and Citi-Walk Sudirman, all in the city center. The areas of shopping malls in Jakarta City have significantly increased from 1.4 million m² in 2000 to 2.4 million m² in 2005 (Tempo, 2006). Likewise, a number of shopping malls have been built in Bandung City, including Bandung Super Mall and Cihampelas Walks. As a result, there has been a great transfer of land ownership from the previous landowners to developers. Uncontrolled land development in the periphery of Jakarta and Bandung City has resulted in many environmental problems, including traffic congestion resulting from heavy flows of commuter traffic between the scattered new towns and the cities, as well as excessive groundwater extraction and air pollution in some industrial in the periphery. It can also be noted that the open green space in Jakarta City has greatly decreased from 28.8% of the total area in 1984 to 9.4% and estimated 6.2% in 2000 and 2007, respectively (Tempo, 2007: 106).

Land-use data in the Bogor–Puncak–Cianjur in the south of Jakarta City (Fig. 1) and in the Bandung Metropolitan Area (BMA) shows that the area of primary and secondary forests, paddy field, garden and estates in the area declined substantially over the period of 1994–2001 (Table 3). In contrast, the land area for industrial activities and settlement increased significantly. During this period nearly 8000 ha and more than 35 000 ha of primary forests in Bopunjur and the BMA have been converted respectively. In other words, more than half of forest areas in Bopunjur and more than one-third of the forest areas in the BMA were converted in only 7 years time. Likewise, more than 4000 ha (16%) and almost 12 500 ha (nearly 20%) of paddy field in Bopunjur and BMA have also been converted. Obviously this conversion has brought significant socioeconomic and environmental impacts to the region. Ironically, the area of Bogor–Puncak–Cianjur has actually been designated as a conservation area, due to its function as a water recharge zone, so that land conversion in the area might result in serious negative environmental consequences in the down stream area, that is, Jakarta City (see Fig. 1).

A number of developers keep the land idle for a period of time for speculation in order to make high profits from rapidly increasing land prices. According to Leaf (1994, 1996), about one-third to one-half of the total area under land development permits in the mid-1990s in Jakarta Metropolitan Region was not actively under development, but was being held off the market by developers.

There have been too many land development permits granted in the past, while the permit-holders were not able to develop fully the sizeable land area that they acquired. In fact, many developers, as permit-holders, simply did not have the financial and technical capabilities. Once the permits are given to the developers, they are allowed to acquire the land within the boundary of permit areas through direct negotiation with the landowners who are often unfairly treated in land transfer, that is to accept unfair compensation. Others are not permitted to buy and develop land within the area, unless they have the permit-holder's consent (see Firman, 2000). Indeed, until very recently, the land development permit system in Indonesia gives an exclusive right to approved developers which in turn may cause speculative land business undertakings.

Land conversion in the Jakarta–Bandung Region (JBR) has also resulted from many violations of land-use plans (rencana tata ruang wilayah) by the local government and private sector in the region, caused by political interests and pressures in placing what are perceived to be profitable economic activities. There are almost no sanctions against the violations, although the plan itself possesses a legal basis. Many development decisions have been made on the basis of proposals submitted by the private sector and developers who have informal and formal access to the authorities, as if the land-use plans are negotiable. In fact, the enforcement is so weak that land-use plans are ineffective in controlling physical development in the region. The Presidential Decree 53/1989 clearly states that industrial estate and new town development should not be located in prime, irrigated agricultural land or on conservation areas, but there have been many violations to the decree due to tremendous increasing demand for the land for industrial estate and new town development in the JBR. Another problem is the local government capacity to control land conversion is inadequate, while pressures from investors are strong.

New town, industrial estate and road development

A number of large-scale residential areas and new towns have been developed in the Jakarta–Bandung Region (JBR) since the early 1980s. According to Leisch (2000) there are three main reasons for developers to build new towns in the region: First, to give them investment opportunities; second, to get huge and quick financial

Table 3

Land-use conversion in the area of Bogor–Puncak–Cianjur (Bopunjur) and Bandung Metropolitan Area (BMA), 1994–2001 (in ha).

Land use	Bopunjur				BMA			
	1994	2001	Change	%	1994	2001	Change	%
Primary forest	12 166.2	5449.4	(6716.8)	(6.76)	57 294.4	55 748.7	(1545.7)	(2.7)
Secondary forest	1205.6	14.0	(1191.6)	(1.20)	39 349.3	5541.9	(33 807.4)	(85.9)
Mining areas	640.8	705.6	64.8	0.07	n.a	n.a	n.a	n.a.
Industrial areas	459.2	647.4	188.2	0.19	2356.2	2478.8	122.5	5.2
Mixed garden	20 499.8	24 809.1	4309.3	(4.34)	42 638.6	85 889.8	43 251.2	101.4
Dry land	6425.0	10 685.0	4260.0	4.29	34 655.6	37 030.7	2375.1	6.9
Grass land	2973.6	2973.6	0.0	0.00	6427.8	6427.8	0.0	0.0
Estates	10 145.3	10 755.5	610.2	0.61	57 680.8	55 946.6	(1734.2)	(3.0)
Settlements	9705.5	12 234.5	2529.0	2.54	29 914.9	33 025.1	3110.2	10.4
Paddy fields	30 328.2	25 939.8	(4388.6)	(4.42)	65 626.1	53 147.4	(12 478.7)	(19.0)
Shrubs	3127.9	3553.0	335.1	0.34	2516.5	3138.5	622.0	24.7
River/lagoon/reservoir	1016.1	1016.1	0.0	0.00	6767.1	6776.6	9.4	0.0
Unused land/open land	593.1	593.1	0.0	0.00	1611.7	1611.7	0.0	0.0
Total areas	99 376.2	99 376.2			346 839.0	346 839.0		

Source: West Java Province Planning Board.

profits; and third, to fulfill people's need for living in a secure, modern and quiet environment. During the 1990s there was easy access for developers to investment funds since both the national and international market encouraged the growth of new town developments and the property sector in general (Dijkgraaf, 2000). In the early 2000s alone there had been about 25 new town development projects in the Jakarta Metropolitan region, ranging from 500 to 6000 ha in size. Nevertheless, by mid-2004 only about three-fifth of the projects were still in operation due to the prolonged economic crisis, including some which occupy 1000 ha or more (Table 4).

The new town development in adjacent areas of large cities in Indonesia during 1980s and 1990s had nothing to do with urban development. It basically reflected the situation of global capitalism, in which transnational capital flowed freely to many economic activities, including the property sector, to sustain and capture its benefits and existence (Douglass, 1998: 4). It can be noted that by end of 1990s the property loan in Indonesia had reached Rp. 73 trillion which made up almost 13% of the total loans of Rp. 574 trillion given by banks in Indonesia (Properti Indonesia, 1998).

By developing new towns the developers had no objectives to decentralize urban socioeconomic activities from Jakarta City to the surrounding areas, as they do not provide working places for residents (see also Leisch, 2000). Therefore, almost all of the new towns have functioned as dormitory towns and are socioeconomically dependent on Jakarta. In fact, new town development in the Jakarta Metropolitan Area (JMA) has intensified the daily interaction

between Jakarta City and the new towns in the region, which in turn makes the traffic problems in the region even worse. Vehicles coming from Bogor, Tangerang and Bekasi contribute significantly to air pollution in Jakarta City everyday.

Realizing this problem, the central government has called upon the Jakarta City government to expand the recently developed busway system to include the adjacent areas of the city, notably Tangerang, Bogor, Depok and Bekasi. It is estimated about 2.5 million private cars, 3.8 motorcycles and 255 000 public transportation vehicles drive along Jakarta City's 7600 km of roads. Moreover, the traffic congestion has significantly resulted in a decrease in average speeds, with most vehicles traveling only 34.5 km/h in 2006, lower than the average speed of 38.3 km/h in 1995. The Jakarta City government is planning to put effort into discouraging people from driving private cars and motorcycles into the city in order to help improve the air quality and to reduce traffic congestion in the city, by the improvement of busway services, increasing parking charges and implementing electronic road pricing (Simamora, 2006a, 2006b).

The primary reasons why a number of middle and high-income groups move to the new towns include avoiding congestion, air pollution and space constraints in the city center, and expecting more reliable security, better infrastructure and living environment, at the same time the location of new towns is still closed to Jakarta city, where most of them work. The new towns in the JMA are built by different developers, resulting in the situation that towns are poorly linked to the existing system of infrastructure in the region and separated from each other. In fact, new town development in the JMA has greatly violated the spatial plan (*Rencana Tata Ruang*) of the area.

Many of the JMA new towns' residents do not want to live in socially and culturally mixed areas for security reasons (see also Leisch, 2000; Somantri, 2007). In fact, many of the new towns in JMA are essentially designed as exclusive residential areas surrounded by walls containing a row and cluster of houses, completely separated from local communities, similar to 'gated communities' just like in Orange Country, Southern California. The physical design of houses in the new towns resembles the ones in developed countries. For example, the new town Royal Sentul (Table 4), located in a hilly area 200–600 m above sea level and designed by Kalges Carter Vail & Partner from California, is equipped with a recreational park, namely 'Taman Fantasia', with facilities for outdoor and indoor games, including mini jets, sky cycles and so on. The houses in this new town are planned and designed according to classical Andalusian and Japanese style (Properti Indonesia, 2002).

Table 4

New town projects occupying 1000 ha or more in Jakarta Metropolitan Region (JBR), 2003.

Project	Area (ha)	Location (Fig. 1)
Bumi Serpong Damai	6000	Tangerang
Cikarang Baru	5400	Bekasi
Kota Tiga Raksa	3000	Tangerang
Lipo Cikarang	3000	Bekasi
Delta Mas	3000	Bekasi
Royal Sentul	2700	Bogor
Kota Legenda	2000	Bekasi
Kota Jaya	1745	Tangerang
Bintaro Jaya	1700	Tangerang
Maharani Citra Pertiwi	1680	Bogor
Resor Danau Lido	1200	Bogor
Citra Raya	1000	Tangerang
Gading Serpong	1000	Tangerang
Kota Baru Parahyangan	n.a.	Bandung

Source: Real Estate Indonesia.

Another example is Lippo Karawaci new town, which was started in early 1990s, equipped with exclusive infrastructure, including a telephone network, cable TV, electricity, water supply, hospitals, hotels, a shopping mall, schools, and cultural and recreational center under an integrated urban management. The developer is a holder of ISO 9001 certificate and has created a western-styled subdivision in the area, namely 'Taman Spain' (Spain Park), which is considered as the masterpiece of Lippo Karawaci in 2000s. There were about 30 000 residents living in this town in 2000 alone (Hogan & Houston, 2001).

The demand for luxury houses in new towns in Jakarta–Bandung Region (JBR) is essentially driven by low-risk investment undertakings, which expect a rise in land and houses prices when resold. In fact, a number of luxury houses in the new towns in the region are bought not for owner occupation but rather for speculative purposes, as land prices increase rapidly.

Many smaller new town developments are also developed in the temperate and hilly Bogor–Puncak–Cianjur (Bopunjur) area, such as Kota Bunga (Flower Town). Many villas are built there with foreign names, such as Villa Orlando, Villa Venezuela and so on, in which most of the houses are only occupied during weekends. It has obviously violated the spatial plan for this designated conservation areas due to its function for water catchment. Housing developments in this area are suspected of having caused severe flooding in Jakarta city in 2002, resulted in the government action of temporarily freezing housing development in the area, however at present it continues (see also Firman, 2004).

The new town developments in the Jakarta Metropolitan Area (JMA) have reinforced spatial segregation in the region in three respects (Firman, 2004): First, it has polarized upper and middle-income groups of JMA residents, creating a several pockets of exclusive new towns and residential areas in which residents enjoyed an exclusive lifestyle, with much better facilities, amenities, infrastructure and securities. The fact is that several new towns in the JMA have become enclaves in themselves; Second, it has enforced social segregation within the new town itself, in which the upper middle-class and lower high-class reside a part of the area which is exclusively designed for the highest security possible. In fact, the developers only invited highly selected people to buy buildings and land in this particular highly exclusive area in the new town (Leisch, 2000). Third, in several new towns in JMA, management of urban development is carried out by the private developers exclusively, instead of by the city government, not permitting people from outside the new town to utilize the facilities. Maintaining the good quality of life in the new town is a primary concern of the company and developers, because it is the most attractive factor for the consumers. Overall, spatial segregation associated with new town development in the Jakarta–Bandung Region (JBR) can be referred as 'voluntary spatial segregation' or 'self segregation' (Fallah, 1996; Greenstein, Sabatini, & Smolka, 2000; UNCHS, 2002), in which the upper and middle-class build housing environment for themselves for the purposes of an exclusive lifestyle and security.

Some of the new towns are also equipped with golf courses, including, Modern Land, Royal Sentul, Bumi Serpong Damai City, Karawaci and Gading Serpong, which has greatly intensified land conversion in the region. By 2000 alone, there were more than 20 golf courses in the Jakarta–Bandung Region (JBR). Among the largest ones is the 27-hole Karawang International Golf Course which occupies 165 ha of land and expected to attract foreign investors and businessmen to work in Jakarta City.

Likewise in the outskirts of Bandung City, almost 3300 ha of paddy field (*sawah*) had been converted into new residential areas in the 1990s (Sarvitri, Rahayu, & Sari, 1997). There are several recent new town developments in the Bandung Metropolitan Area (BMA), including 'Kota Baru Parahyangan', 'Kota Bunga' and 'Estat

Batununggal Indah'. By far and the largest is 'Kota Baru Parahyangan' located near the tollgate Padalarang on the way towards Jakarta City (Table 2). This new town won the prestigious award of 'The Best Housing Development Project in Indonesia, 2007' (*Anugerah Adhika Niwasthana*) for the category of area larger than 200 ha from Real Estate Indonesia, covers a planned area of 1250 ha to be developed for 20 years started from 2000, and is designed for clusters of residential areas of about 15 ha consisting of about 500 houses and is developed to become a self-contained new town with some exclusive primary to high schools and even a university in the near future. 'Kota Baru Parahyangan' is a rather unusual new town, since most of the new town projects of this size are built in the JMA, instead of in the BMA. Just like in the JMA, the new town developments in the BMA also tend to create a spatial segregation in the region, albeit at a smaller scale.

Industrial estates in the Jakarta Metropolitan Region (JBR) occupied a total land area of nearly 11 000 ha by the end of 2005, approximately four-tenths of which are located in Bekasi District, one of the largest concentrations of manufacturing activities in Indonesia (Collier International, 2005), including the Lippo City Industrial Park, Cikarang Industrial Estate, East Jakarta Industrial Park, Bekasi International Industrial Estate and MM 2100 Industrial Estates (Table 5). Another one-third of these industrial estates are located in Karawang District, including Karawang International Industrial Estates, Kawasan Industri Kujang Cikampek, Kawasan Industri Mitra Karawang, Mandalapratama Permai Industrial Estates, Suryacipta City of Industry and Taman Niaga Karawang Prima (Table 5). These industrial estates have a strong market demand due to their proximity and easy access to Jakarta City. Nevertheless, more recently foreign investors are still holding back new industrial investment because of several reasons, notably

Table 5
Industrial estates in Jakarta Metropolitan Region, 2005.

Industrial estate	Planned area (Ha)	Location (district)	Remarks
Jababeka	3000	Bekasi	Town and industrial estate
Bekasi 3000 Industrial Estate	250	Bekasi	
East Jakarta Industrial Park	320	Bekasi	
Kawasan Industri Gobel	52	Bekasi	
Bekasi International Industrial Estate	200	Bekasi	
Cikarang Industrial Estate	1330	Bekasi	
Lippo Cikarang Industrial Estate	1000	Bekasi	
MM 2100 Industrial Town	1000	Bekasi	
Patria Manunggal Industrial Estate	220	Bekasi	
YKK Industrial Estate	20	Bekasi	
Delta Silicon-Lippo Cikarang	500	Bekasi	In plan
Karawang International Industrial Estate	1118	Karawang	
Kawasan Industri Kujang Cikampek	140	Karawang	
Kawasan Industri Mitra Karawang	434	Karawang	
Mandalapratama Permai Industrial Estate	292	Karawang	
Suryacipta City of Industry	1473	Karawang	
Taman Niaga Karawang Prima	200	Karawang	
Kota Bukit Indah Industrial City	1300	Purwakarta	
Cibinong Centre of Industrial Estate	32	Bogor	
Sentul Industrial Estate	5	Bogor	In plan
Modern Cikande	80	Serang	In plan
Cilandak Commercial Estate	11	South Jakarta	
Jakarta Industrial Estate	570	East Jakarta	
PuloGadung			
Pluit Distribution Center	80	North Jakarta	

Source: Collier International, 2005; Investment Coordinating Body, 2005.

socioeconomic and political uncertainties, uncompetitive taxes, and discouraging labor relations.

The demand for industrial land in the Jakarta Metropolitan Area (JMA) has greatly increased due to development of direct foreign and domestic investment in the region. In total, the occupancy rate of industrial estates in this region reached 70%. It is estimated that by mid-1990s about half of the total foreign and domestic investment in Indonesia, excluding oil and gases, were located in the area (Firman, 1999). The cumulative approved direct foreign investment in Indonesia 2000–2004 had reached US \$ 64 803.5 million, nearly 58% (US \$ 37 112.8 million) of which was located in the JMA (Central Board of Statistics, 2006). Meanwhile, the cumulative approved domestic investment in Indonesia had reached Rp. 265 176.1 million over the same period, nearly one-third (about Rp. 82 342 million) of which were invested in the JMA (Central Board of Statistics, 2006). In short, the JBR development has been due to market forces, especially new town and industrial estate development. These reflect the importance of JMA in the national economy.

Thus far there is no large industrial estate in the Bandung Metropolitan Region, but there are at least there are two industrial zones, which are smaller in size compared to most industrial estates in the JMA, including the Cibaduyut industrial zone which specializes in footwear products, and Gedeage.

The Jakarta–Bandung belt is one of the most crowded transport corridors in Java along with the belt of North Shore, served by railway, road and air mode of transportation. This corridor is estimated to serve more than 78 million passengers in 2004, mostly traveled from Bandung to Jakarta and vice versa for businesses and other purposes, about 95% of which served by road mode (Lubis, Dharmowijoyo, & Armijaya, 2005: 76). Meanwhile, the total number of registered passenger cars, trucks, buses, and motorcycles in Jakarta City increased by about 10.6% per year over the period of 1993–2001, and it was estimated that twice as many people would be using some form of vehicles by 2020 (Hata, 2003: 38). It can also be noted that the JMA's railways carried about 118 million passengers which cover nearly 3600 million passenger-km in 2002. The JMA's passenger density reached 58 000 passengers/day/km in 2002, which is six times the figure for Indonesia's as a whole (Hata, 2003: 44).

The above transport situation in the JMA indicates the immediate need to develop a railway system in the area to cope with the problems resulting from overdependence on road transport. Realizing this problem, the government has been implementing the integrated Transportation Master Plan for the JMA (SITRAMP) which is expected to establish a comprehensive rail and road transport system with the assistance from the Japanese government (Hata, 2003).

Jakarta and Bandung City are also connected by several toll roads, including the Padalarang–Purwakarta–Cikampek road and the Jakarta–Ciawi Road. The state-owned toll road company (PT Jasa Marga) has recently completed the Purwakarta–Padalarang toll road construction, which makes the travel time between two cities shortened from three to only 2 h. As Lubis et al. (2005) argue that the rail mode of transportation will lose its share, because more passengers will shift to road mode, including private cars and buses as the road mode allows them to travel with a shorter time than the rail. Another toll road connecting Jakarta Tanjung Priok International Seaport with Cikarang in Kabupaten Karawang, about 34 km long, is now being planned with Rp. 2.4 trillion investment.

The opening new toll road Jakarta–Bandung has made people from Jakarta to travel to Bandung easily during the weekend for shopping and recreation, which in turn has encouraged several economic activities to grow in Bandung, including hotels, restaurants, food stores, and garment factory outlets. In fact, Bandung is becoming a tourist city, reflected in the city own income (*Pendapatan Asli Daerah: PAD*) from tourism sector which reached about

40% of the total PAD Rp. 234 billion in 2006 with rate of growth at 9% per year since 2004 (Kompas, 2007). This development has also triggered new socioeconomic development in some side areas along the toll road, including rest areas and restaurants. However, it has also worsened the traffic congestions in the city during weekend.

Early reform and decentralization era

Under the pressures of justice and democratization issues, Indonesia's Parliament passed Law 22/1999 and 25/1999 concerning the regional autonomy and fiscal decentralization in May 1999. These two pieces of legislation are basically aimed primarily at curbing separatist sentiment in the outlying provinces of Indonesia in order to avoid a break up Indonesia into several small tiny countries, and to stop the exploitation of central government of local governments. Moreover, the legislations are also expected to bring the government closer to the people and to empower the local and provincial government, local legislature councils and local communities (Brodjonegoro, 2006; Firman, 2003). The other objectives are to make use public funds in a more effective and efficient manner in line with local development needs and to improve the quality of public service provision (see also Booth, 1999; Rasyid, 2004).

Decentralization is essentially a transfer of a significant degree of responsibility and authority for public revenues and expenditures from the central government to the local government under the principle of as much as autonomy as possible and as much as central power as necessary (Alm, Aten, & Bahl, 2001: 84; Koswara, 2002: 39). Nevertheless, decentralization is needed for the effective and efficient provision of infrastructure and public services for urban and regional development in Indonesia, because until very recently this task had been largely the responsibility of the central government which was ineffective and inefficient and did not necessarily meet the local needs in a balanced way (Firman, 2002, 2003; Rondinelli, 1990). According to Fritzen and Lim (2006: 1) in general decentralization is expected to improve levels of public participation, accountability, administrative efficiency and responsiveness to the local needs.

The legislation was then amended in 2004 became Law 34/2004 and 35/2004 but this did not change the main objectives. The only item added in the law was about the direct election (*Pilkada Langsung*) for head of the provinces, districts, and municipalities. It is believed that wider local autonomy may spur local development, as it may encourage the local government and communities to take initiative for necessary action and regulation for their own benefits.

Before the year 2000, especially during President Soeharto's era from 1966 to 1998, the central government of Indonesia concentrated decision-making at top and did not give much decision-making authority to the local level, while the later were supposed only to implement the central government's decisions, including the implementation of service delivery at the local level, which was considered as decentralization (Crane, 1995; Devas, 1997; see also Fritzen & Lim, 2006). Moreover, the dominance of the central government in this function weakened the local authority to provide infrastructure and services to local communities (Firman, 2003; see also Rondinelli, 1990: 6–7). According to Cheema and Rondinelli (1983: 296) a decentralization policy can only be successfully implemented if it is clearly defined and if its scope is limited to reallocating functions among the central government units.

By 2008 the progress of decentralization implementation has been patchy, in which some local governments have performed better in fulfilling local development management, notably in public service delivery, while some others have done it even worse as officials are involved in corruption and bribery. As Hadiz (2004: 711)

maintains, decentralization in Indonesia has been marked by the rise of new pattern of highly decentralized corruption practiced by local officials, money politics, and political gangsters (see also [Fritzen & Lim, 2006](#): 715; [Kuncoro, 2002](#)), and instead of empowering the most disadvantaged parts of civil society, the decentralization has greatly taken off without them. As a result, the progress of decentralization implementation in Indonesia has been very slow.

Not all local governments are ready to implement decentralized urban and regional development, because at present most of them simply do not have enough institutional, financial and technical capacity for it (see also [Alm et al., 2001](#)). In fact, many local governments are still in the euphoria condition, thinking of their local region as a 'kingdom' of their own and of resources available as their exclusive possessions, but they are not really concerned with knowing how to use it for regional development purposes.

Both the new laws of regional autonomy and fiscal decentralization do not provide the local government with any significant control over taxation, while the Indonesia's taxation law reserves greater control to the central government on tax-raising, limiting the local government from collecting significant taxes and charges. As a result, local government taxing powers are weak, which has resulted in the low capacity of local government (see [Firman, 2003](#)). At present there is a tendency for local government to maximize their authority to generate local own income (*Pendapatan Asli Daerah*) without giving consideration to the socioeconomic and political conditions of the region in which they are part of. On the contrary, many local governments try to build infrastructure, including seaports and airports, to compete with other regions and cities, disregarding the effectiveness and efficiency of such huge investments. Obviously, the situation has in turn brought disadvantages for both their own Kabupaten and Kota as well their neighboring areas.

The local governments of Kabupaten and Kota in the Jakarta–Bandung Region (JBR) are no exception. There is a tendency to try to exploit local resources, including land, water and other physical assets to maximize their own income (PAD). The local governments of Kabupaten Bogor and Cianjur, for instance, continue to allow developers to build and develop the temperate and beautiful scenic area of Puncak Strip (Jalur Puncak) intensively for tourist resort villas and residential areas for middle and high-income groups from Jakarta, although this area has been designated as a conservation area due to its function for water recharging.

The local governments indeed receive some financial benefits from development permits granted to the developers, but this uncontrolled development in turn might bring negative environmental consequences for the downstream areas, notably Jakarta City, including water scarcity in the dry season and floods in the rainy season. Likewise, the developing residential areas in northern Bandung for middle- to high-class residential areas could have serious negative impact on Bandung City in the near future, since northern Bandung is found to be a water recharge area for the BMA. Many Non-Government Organizations and communities in Bandung have protested this uncontrolled development, but the local government and developers seem not to care about it.

The local governments receive their total amount of grant from the central government based on the 'fiscal gap', that is, the difference between estimated expenditure and revenue capacity, which is called General Allocation Funds (DAU) aimed to help reduce central-local financial imbalances. The DAU is basically an unconditional block grant given to the local government which can be used according to local needs at the discretion of local government's authority.

Under the legislation of Fiscal Decentralization the annual local government revenues consist of an equalization grant from the General Allocation Funds (Dana Alokasi Umum – DAU) to substitute the central government grants, most notably Presidential Decree

Grants, in the past; natural resources utilization revenues, notably fisheries, forest products and oil and gases; revenue sharing income from land and building taxes and employee taxes; own income (Pendapatan Asli Daerah – PAD); and a special allocation fund (Dana Alokasi Khusus – DAK) to be granted by the central government to any districts on the basis of special need.

All the local governments in the Jakarta–Bandung Region (JBR) have received an increasing amount of the General Allocation Fund (DAU) since 2001 ([Table 6](#)). The fact is that Kabupaten Bandung has earned the largest DAU among local governments in Indonesia, since 2001. Therefore, the local government now have much more financial resources to spend at their own discretion than in the past.

Many local governments in the JBR have a negative parochialism attitude, resulting in a number of problems in service provision which need cross-border cooperation, including solid waste management and water supply. The District of Bekasi which borders with Jakarta City, has imposed high charges to the Jakarta City government for using the dump site in their area, causing problems of finding dump sites for Jakartans. A similar problem of waste disposal also has happened between Bandung City and Bandung District in the BMA, resulted in problem of mounting garbage in the city recently.

The Jakarta–Bandung Region (JBR) is a compact urban region that should be managed as an inseparable region through an integrated spatial development plan (Rencana Tata Ruang), to ensure sustainable development of this region although it consists of some Kabupaten and Kota which have their own authority. In the regional autonomy and decentralization era, this seems to be a much more complicated matter and difficult to do. What is badly needed at present is the good leadership of local government leaders, including in West Java, Banten and Jakarta Provincial government as well as the central government. They should have the vision and awareness that the region should be developed through a cooperation of all stakeholders and make coordination work for themselves. Institutional development for the JBR is very critical. As [McGee \(2005: 51\)](#) argues, Asian countries, including Indonesia, need to develop their own national and local urban development policies, strategies, and institutions which are indigenously formulated, including the institutional set up (see also [Friedmann, 1999](#); [Laquian, 2005](#)).

Former Jakarta's Governor General Ali Sadikin proposed regional development of Jakarta and the surrounding areas in the early 1970s, but the recent Jakarta's Governor Sutiyoso recently stressed the need for development of integrated "Megapolitan Jabodetabekjur" comprising Jakarta City, Bogor City, Depok City, Bekasi City, Tangerang City, Bogor District, Bekasi District, Tangerang District and Cianjur District as a solution to regional development problems in the areas. Many parties from the neighboring

Table 6

General allocation fund (DAU) and product domestic regional bruto (PDRB) of local government in Jakarta–Bandung Region (JBR), 2001–2006 (Billion Rupiah).

Local Government	DAU			PDRB
	2001	2003	2006	2000
Jakarta City	773.0	734.9	768.1	189 075.4
Bekasi City	211.6	256.5	420.3	8875.6
Bogor City	168.7	197.4	302.5	2671.6
Depok City	167.2	209.5	313.2	3489.1
Bandung City	385.2	416.7	632.4	14 423.0
Bandung District	654.8	726.2	1168.6	19 978.1
Bekasi District	235.4	231.2	284.9	30 267.2
Bogor District	479.6	517.1	807.6	10 597.8
Cianjur District	335.6	389.8	675.9	5221.6
Tangerang District	313.1	367.0	508.0	12 196.8
Karawang District	220.0	328.0	532.1	7314.9

Source: Ministry of Finance and Central Board of Statistics.

Provinces accuse Sutiyoso of having a political ambition to annex the region of Bogor–Tangerang–Bekasi–Cianjur to Jakarta City and for himself to become a Governor or a Minister in charge with development of the JMA (Jabodetabekjur). He could have such an ambition, but the idea of “Megapolitan Jabodetabekjur” could be an alternative solution to development in the JMA.

Concluding remarks

The Jakarta–Bandung Region (JBR) which consists of The Jakarta Metropolitan Area (JMA) and the Bandung Metropolitan Area (BMA) play very important role in the national economy. Its physical development has formed an urban belt form Jakarta City to Bandung City of about 200 km, reflecting a formation of mega-urban region characterized by a mixture of rural and urban activities and a blurred rural–urban distinction in the Jakarta–Bandung Region (JBR). Population growth in the urban centers, notably Jakarta and Bandung City, have significantly declined, whereas in the adjacent areas population growth is relatively high, which may reflect the rapid spillover of Jakarta and Bandung City to the surrounding areas. In fact, a large number of many former residents in neighborhood within the Jakarta City center have moved to adjacent areas. The increase in number and percentage of urban localities (*desa urban*) in the JBR also shows the urban transformation in the region.

Although the economic crisis in the late 1990s and early 2000s saw a significant slow down of the JBR's development due to a decline in demand for land for industrial estates and new town developments in the region, the JBR has been developing rapidly in the mid-2000s as it was before the economic crisis. Overall, the JMA and the BMA are experiencing a transformation from single-core to multi-core cities, greatly driven by the capitalistic private sectors, including industrial manufacturing, trade and services, and property enterprises, which might have taken great advantages out of this opportunities.

In the past three decades the development of socioeconomic activities in the JBR has resulted in several environmental problems, most notably traffic congestion due to heavy flows of commuters traffic between Jakarta and its surrounding areas, air pollution in some industrial areas, and excessive groundwater extraction. The JBR also faces the problem of uncontrolled land conversion resulting from many violations of land-use plans by local governments and the private sector, caused by profitable and political interests.

A number of new towns and large-scale residential areas have been developed in the JBR since the 1980s. However, the new towns in JBR are socioeconomically dependent on Jakarta City and Bandung City and have functioned as ‘dormitory town’, instead of self-contained towns. Many of the new towns are designed as exclusive residential areas resembling the ‘gated communities’ in developed countries. In fact, new town development, especially in the Jakarta Metropolitan Area (JMA), have reinforced spatial segregation by life style and income status in the area. Likewise, industrial estates are also growing in the JMA due to development of domestic and direct foreign investment in the region for proximity and easy access to Jakarta City. However, many low-income residential areas in the city center have been removed and converted to several commercial areas.

Jakarta and Bandung City are now connected by several new toll roads constructed since early 2000. It has made large number of Jakarta's middle and upper-class residents travel to Bandung City during the weekend, for shopping and recreation, and in turn stimulated development of economic activities in the city, including restaurants, hotels, and garment factory outlets, although it has also made the traffic problem worsen.

The new regional autonomy and fiscal decentralization policy makes urban development in the Jakarta–Bandung Region (JBR) become local authority affairs, which implies that it has to be planned and implemented according to the local needs by the local authority and communities, while in the past it was strictly under the central government control and direction. However, as in other Kabupaten (District) and Kota (Municipality) in Indonesia, there is a tendency of the local governments in the JBR to try to exploit local resources excessively, including water and land to maximize their own income. Moreover, the parochialism attitude of many local governments has caused a number of problems in services which need cross-border cooperation, including water supply and waste management.

The mega-urbanization process in the JBR is continuing, even at a faster rate than that in the period of the 1980s to mid-1990. However, the new regional autonomy and fiscal decentralization policies tend to make the region fragmented in terms of an integrated urban region, as each of the local government in JBR has their own discretion to manage their territory, which might potentially result in excessive resource utilization in the region.

Overall, this study has tried to examine the continuity and change of the mega-urbanization in the Jakarta–Bandung Region, the largest concentration of urban population in Indonesia. Admittedly, however, this study is largely descriptive in character, but it is one among a few studies which examines on the recent development of the Jakarta–Bandung Region development.

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